

RAFFLES INSTITUTION
2018 YEAR 6 PRELIMINARY EXAMINATION

Higher 1



CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

8873/02

Paper 2 Structured Questions

11 SEPTEMBER 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not open this question booklet until you are told to do so.

Write your name, class and index number in the spaces provided at the top of this page.
Write in dark blue or black pen in the spaces provided.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

For Examiner's Use		
Section A	1	/ 13
	2	/ 5
	3	/ 20
	4	/ 11
	5	/ 11
Section B (Please circle the question you have attempted)	6	/ 20
	7	/ 20
Total	/ 80	

A Data Booklet is provided. Do not write anything in it.

You are reminded of the need for good English and clear presentation in your answers.

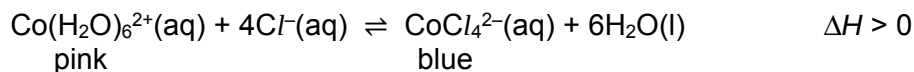
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **22** printed pages.

Section A

For
examiner's
useAnswer **all** questions in this section, in the spaces provided.

- 1 (a) A student performed an experiment which involved two complexes of Co(II) ions and their colours are given below.



- (i) Write the K_c expression for this reaction and state its units.

units[2]

- (ii) Solution **X** was prepared in which the initial concentrations of $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and Cl^- ions were 1.80 mol dm^{-3} and 4.00 mol dm^{-3} respectively. After equilibrium was reached, the concentration of CoCl_4^{2-} ions was 0.92 mol dm^{-3} .

Calculate the value of K_c for this equilibrium.value of K_c [2]

- (iii) Two experiments were conducted using solution **X** and some experimental observations and deductions were given in Table 1.1.

Table 1.1

test	observations	equilibrium position	predominant ions present
① Test-tubes 1 and 2 contain solution X .	violet solution	near the centre	
② Place test-tube 2 in a beaker of hot water.	violet solution turns blue		

Complete Table 1.1.

[2]

- (iv) Explain the observations in test ② using Le Chatelier's Principle.

For
examiner's
use

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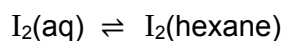
.....

.....

.....[2]

- (b) Use of the Data Booklet is relevant to this question.

The same student then performed a different experiment to investigate the solubility of iodine in aqueous solution and hexane.



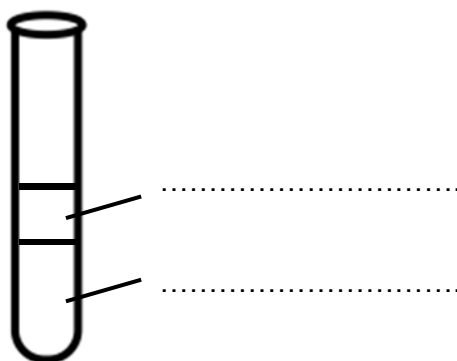
- (i) State the colour of iodine in

aqueous solution

hexane [1]

- (ii) Hexane was added to a test-tube containing iodine dissolved in water and the mixture was shaken well for 10 seconds and allowed to settle. Two immiscible layers were observed.

Given that the densities of hexane and water are 0.655 g cm^{-3} and 1.00 g cm^{-3} respectively, label the colour of the two layers in this test-tube and explain your answer.



explanation

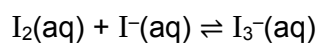
.....[1]

- (iii) Explain why iodine is more soluble in hexane than in water.

For
examiner's
use

.....
.....
.....
.....[2]

- (iv) Iodine can form the tri-iodide ion, I_3^- , with potassium iodide according to the reaction as shown.



The tri-iodide ion is soluble in water. Explain why this is so.

.....
.....[1]

[Total: 13]

- 2 Poly(ethene) exists in two different forms, namely the low density poly(ethene) (LDPE) which has lots of side chains, and the high density poly(ethene) (HDPE), which has fewer and shorter side chains.

- (a) Explain, with the aid of suitable diagrams, why the presence of fewer and shorter side chains results in high density poly(ethene).

LDPE	HDPE

explanation

.....

.....

.....[2]

- (b) The manufacture of Lego bricks uses polymers which are commonly made from crude oil. These polymers are inert, very durable and less likely to fade in colour.

Explain why the disposal of such polymers is an environmental concern and suggest a strategy to overcome this problem.

explanation

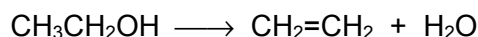
.....

strategy

.....[2]

- (c) Much research has been done to promote the use of alternative starting materials instead of crude oil. Recently, Lego has started using poly(ethene) derived from sugar cane.

The ethanol in sugar cane is converted to ethene as shown in the reaction below.



Name the type of reaction.

.....[1]

[Total: 5]

- 3 (a) The element oxygen can exist as a number of isotopes.

For
examiner's
use

Table 3.1

isotopic species	protons	neutrons	electrons	electronic configuration
$^{16}_8\text{O}$				$1s^2$
		10	8	$1s^2$

Complete Table 3.1.

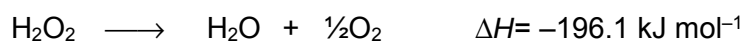
[2]

- (b) The Haber-Weiss reaction generates highly reactive free radicals from H_2O_2 .

A free radical is formed when the O–O covalent bond in H_2O_2 breaks. Explain, with the aid of a 'dot-and-cross' diagram, why the free radical is highly reactive.

explanation[2]

- (c) Hydrogen peroxide can undergo disproportionation reaction in the presence of the Fe^{2+} catalyst.



The uncatalysed reaction has an activation energy of $+49 \text{ kJ mol}^{-1}$.

- (i) Using the data provided, draw a labelled energy profile diagram for the uncatalysed reaction on Fig 3.2.



Fig 3.2

[2]

- (ii) Explain what is meant by the term *activation energy*.

For
examiner's
use

.....
.....[1]

- (iii) With the aid of a Boltzmann distribution curve, explain the effect of a catalyst on the rate of reaction.

explanation

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.....

.....

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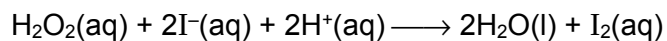
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.....

.....[3]

- (d) The Harcourt and Esson reaction involves hydrogen peroxide and acidified potassium iodide.

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use



The rate of reaction can be followed by measuring the concentration of H_2O_2 remaining at various times. The following table shows how $[\text{H}_2\text{O}_2]$ varies as the reaction proceeds.

time / s	0	100	200	300	400	500	600	700
$[\text{H}_2\text{O}_2] / \text{mol dm}^{-3}$	0.030	0.024	0.019	0.015	0.012	0.010	0.0080	0.0060

- (i) Plot a graph of $[\text{H}_2\text{O}_2]$ against time on Fig 3.1.

[1]

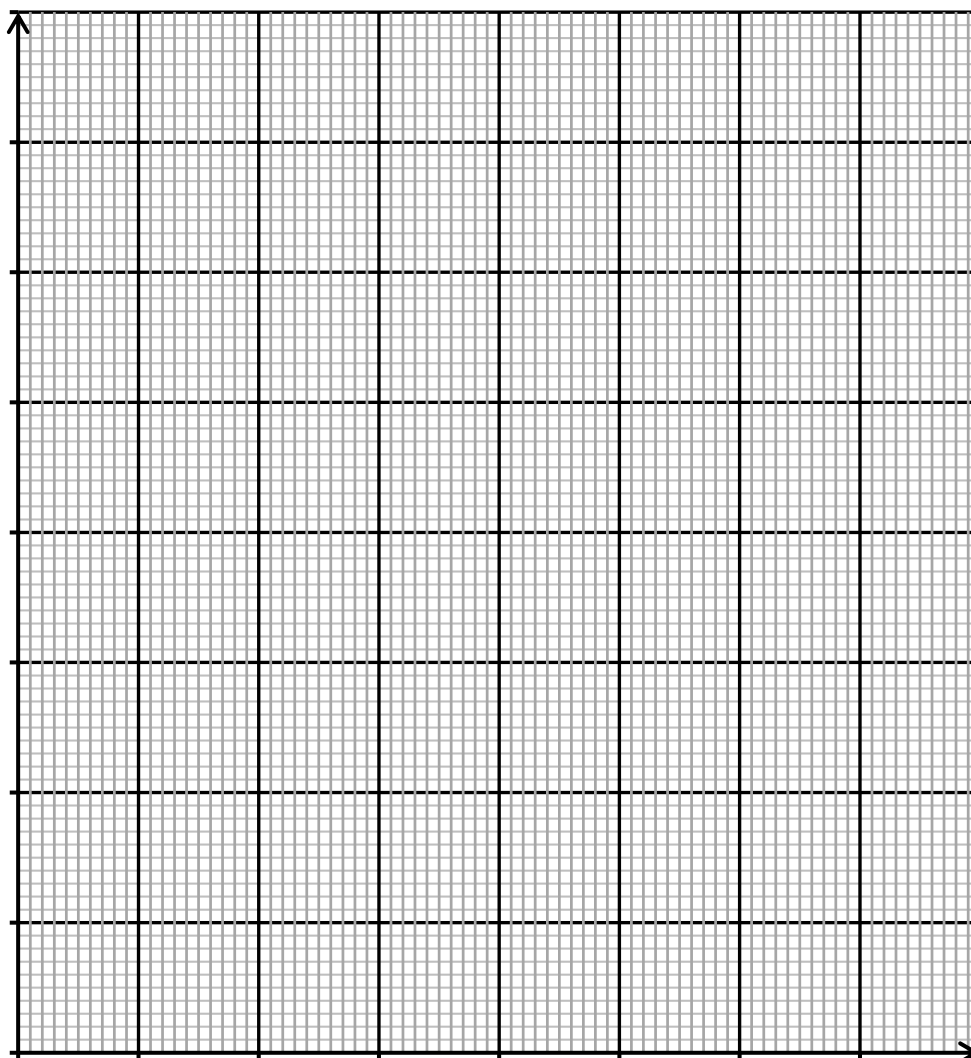


Fig 3.1

Use Fig 3.1 for **(d)(ii)** and **(d)(iii)**. By showing all your working and drawing clearly any construction lines on your graph, determine

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examiner's
use

(ii) the order of reaction with respect to $[H_2O_2]$,

order of reaction [2]

(iii) the initial rate, in $\text{mol dm}^{-3} \text{s}^{-1}$.

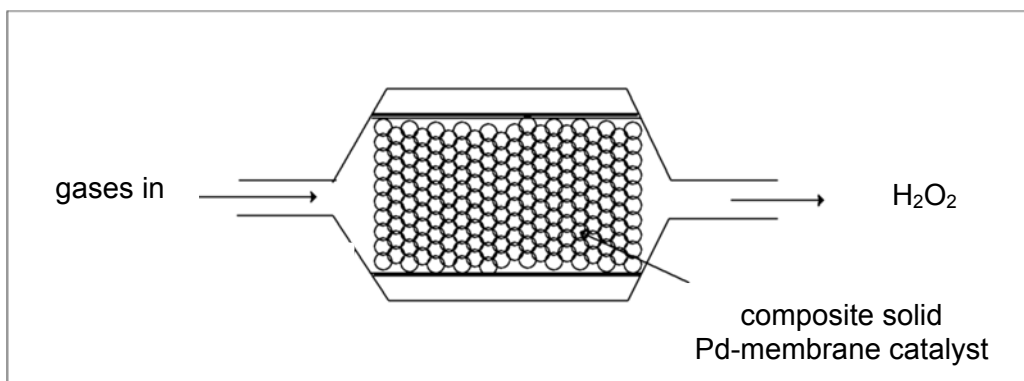
initial rate $\text{mol dm}^{-3} \text{s}^{-1}$ [1]

- (e) H_2O_2 is in high demand in chemical industries as it is considered an environmentally-friendly oxidising agent.

For
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Industries have been finding ways to produce H_2O_2 in a more economical manner by direct oxidation of H_2 with O_2 .

In recent times, a novel composite solid palladium membrane catalyst for the direct oxidation of H_2 to H_2O_2 has been used.



- (i) Explain why it is environmentally-friendly to use H_2O_2 as an oxidising agent.

.....[1]

- (ii) Write a balanced equation for the direct oxidation of H_2 to H_2O_2 .

.....[1]

- (iii) State the type of catalyst involved in the direct oxidation of H_2 to H_2O_2 .
Describe how this type of catalyst speeds up the gaseous reaction.

type of catalyst

.....

[4]

[Total: 20]

- Include the approximate pH value of any resulting solutions and write equations for any reactions that occur.

(ii) Suggest what influence the type of bonding present in the chlorides of magnesium and phosphorus has on their reaction with water.

.....[2]

- (b) Unlike aluminium, boron is a non-metal and share more similarities with carbon and silicon than with the other Group 13 elements.

- (i) Like SiCl_4 , BCl_3 undergoes hydrolysis to form compound **M** and white fumes.

Suggest the formula of compound **M** and write an equation for this reaction.

compound **M**

equation[1]

- (ii) Similar to carbon, boron nitride, BN, exists in a diamond-like form and in a graphite-like form.

In the diamond-like form, each nitrogen atom is bonded to four boron atoms, and each boron atom to four nitrogen atoms.

In the graphite-like form, each layer of BN consists of alternating boron and nitrogen atoms.

In the boxes provided below, draw the diamond-like and graphite-like structures of BN and predict a different physical property for each structure.

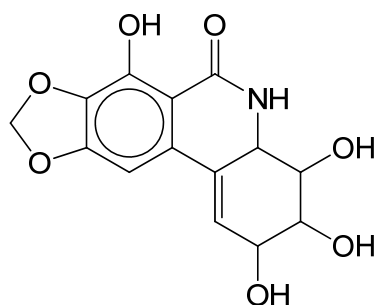
	diamond-like form	graphite-like form
structure		
physical property		

[4]

[Total: 11]

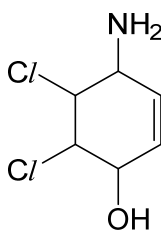
- 5 Daffodils are bright yellow flowers which bloom in spring. When placed in a vase together with other flowers, they can cause premature wilting and death of these flowers. This is due to the toxic alkaloid compounds found in the daffodil mucilage. An example of the toxic alkaloid compounds found in daffodils is narciclasine.

For
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use



narciclasine

Compound **P** is used in the synthesis of narciclasine.



P

- (a) State the functional groups present in **P**.

.....[3]

- (b) Use the table of characteristic infra-red absorption frequencies in the *Data Booklet* to answer this question.

Infra-red absorptions can be used to identify functional groups in organic compounds. For example, carbonyl compounds show absorption at $1670 - 1740 \text{ cm}^{-1}$.

Use the table to identify an infra-red absorption range that will be shown by

- (i) narciclasine but not by **P**

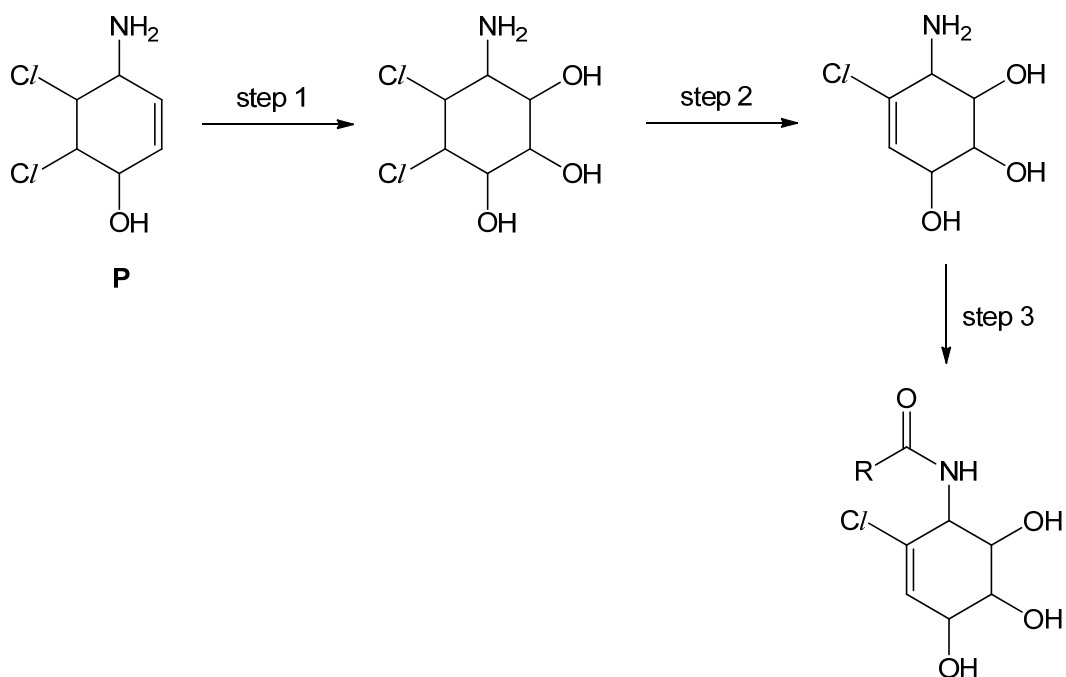
..... cm^{-1}

- (ii) **P** but not by narciclasine

..... cm^{-1}

[2]

The following shows part of the synthesis of narciclasine from compound **P**.



- (c) (i) Step 1 involves the oxidation of **P**.
Write a balanced equation for this reaction using the [O] notation.

.....[1]

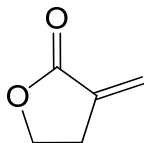
- (ii) State the reagents and conditions for steps 2 and 3.

step 2:

step 3:[2]

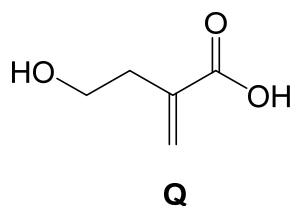
- (d) Tulip is one of the flowers with reduced vase life when placed together with daffodils. When handling tulips, skin contact with the tulip bulbs should be avoided to prevent 'tulips fingers', which is an allergic reaction caused by tulipalin A.

For
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use



tulipalin A

Tulipalin A can be formed from compound **Q**.



Q

- (i) State the type of reaction and the reagents and conditions required.

type of reaction

reagents and conditions[2]

- (ii) Draw the displayed formula of the product formed when **Q** is heated with acidified aqueous $K_2Cr_2O_7$.

[1]

[Total: 11]

Section B

Answer **one** question from this section, in the spaces provided.

For
examiner's
use

6 (a) Iron is widely used in the manufacture of steel.

(i) Describe the structure and bonding in iron. Draw a diagram to illustrate your answer.

.....

[2]

(ii) By considering your answer to (a)(i), state **two** physical properties that you would expect iron to possess. Explain why it possesses these properties.

property

explanation

.....

property

explanation

.....[2]

- (b) Anaemia is a condition caused by a lack of iron in the body.

Iron tablets containing iron(II) sulfate, FeSO_4 , are administered to patients suffering from anaemia. The amount of iron(II) sulfate in the tablets can be determined titrimetrically using acidified potassium manganate(VII), KMnO_4 .

In an experiment, five iron tablets were crushed and dissolved in dilute sulfuric acid. This solution was made up to 250 cm^3 with deionised water.

25.0 cm^3 of the diluted solution required 27.55 cm^3 of $0.005 \text{ mol dm}^{-3}$ acidified potassium manganate(VII) for complete reaction.

- (i) Write a balanced equation for the overall reaction between Fe^{2+} and MnO_4^- .

.....[1]

- (ii) Calculate the amount, in moles, of iron(II) sulfate present in the original 250 cm^3 solution.

[3]

- (iii) Hence, calculate the mass of iron(II) sulfate present in one tablet.

[2]

- (iv) A student decided to repeat the experiment, but dissolved the crushed tablets in hydrochloric acid instead.

State, with reasoning, what effect this will have on the volume of acidified KMnO_4 required for complete reaction and on the calculated mass of FeSO_4 in one tablet.

.....
.....
.....
.....[2]

(c) The melting points of GeO_2 and SeO_3 are 1115°C and 118°C respectively.

- (i) By considering the melting points of the oxides and the positions of Ge and Se in the Periodic Table, suggest the structure and type of bonding present in each of the oxides.

.....

[2]

- (ii) SeO_3 can react with either sodium hydroxide or hydrochloric acid. Write a balanced equation for its reaction with either hydrochloric acid or sodium hydroxide.

.....[1]

(d) Ethanol can be used as a fuel. Most existing car engines can run on blends of 15% ethanol and 85% gasoline.

Ethanol is commonly produced by the fermentation of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.

- (i) Calculate the average oxidation number of carbon in glucose.

[1]

- (ii) Write a balanced equation for the disproportionation reaction of glucose to produce ethanol and carbon dioxide only, by using oxidation numbers or otherwise.

.....[1]

(e) Gasoline contains a mixture of alkanes, including octane, $\text{CH}_3(\text{CH}_2)_6\text{CH}_3$.

- (i) Write a balanced equation, including state symbols, for the complete combustion of liquid octane under standard conditions.

.....[1]

- (ii) Some relevant standard enthalpy change of formation values, $\Delta H_f^\ominus$, are given in Table 6.1. Using your answer to (e)(i) and the data in Table 6.1, calculate the standard enthalpy change of combustion of liquid octane.

For
examiner's
use

Table 6.1

compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{CH}_3(\text{CH}_2)_6\text{CH}_3(\text{l})$	-249.7
$\text{CO}_2(\text{g})$	-393.5
$\text{H}_2\text{O}(\text{l})$	-285.8

standard enthalpy change of combustion of octane.....[2]

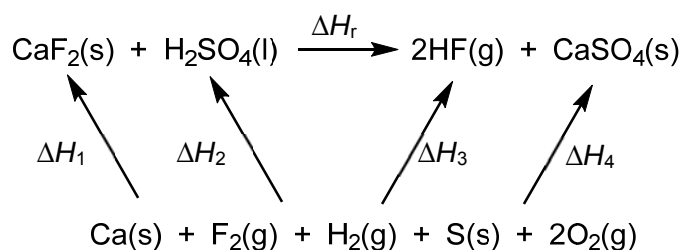
[Total: 20]

- 7 (a) Hydrofluoric acid, HF, can be manufactured by treating calcium fluoride, CaF_2 , with concentrated sulfuric acid, H_2SO_4 .

- (i) Draw a 'dot-and-cross' diagram to show the bonding in H_2SO_4 . State the shape of the molecule around the S atom.

shape of molecule around the S atom[2]

- (ii) The diagram below shows an energy cycle involving the reaction of calcium fluoride with concentrated sulfuric acid.



$$\Delta H_1 = -1220 \text{ kJ mol}^{-1}$$

$$\Delta H_2 = -814 \text{ kJ mol}^{-1}$$

$$\Delta H_3 = -542 \text{ kJ mol}^{-1}$$

$$\Delta H_4 = -1434 \text{ kJ mol}^{-1}$$

Using the energy cycle and data provided, calculate the

- ① enthalpy change of formation of gaseous hydrogen fluoride, ΔH_f ,
- ② enthalpy change of the reaction, ΔH_r .

ΔH_f

ΔH_r [3]

(b) Compared to other hydrogen halides, HF shows significantly different chemical and physical properties.

(i) By quoting relevant values from the *Data Booklet*, describe and explain the trend in thermal stability of the hydrogen halides HF, HCl and HBr.

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.....[3]

(iii) Calculate the pH of a $0.050 \text{ mol dm}^{-3}$ solution of HCl.

[1]

(ii) With the aid of an equation, briefly explain why HF is a weak acid.

.....

.....

.....[2]

(iv) A student proposed diluting a solution of HF so as to

- ① increase its acid dissociation constant, K_a and
- ② lower its pH value.

Explain if the student is correct.

.....

.....

.....

.....[2]

- (c) To standardise a solution of aqueous ammonia, a titration was carried out with $0.050 \text{ mol dm}^{-3}$ solution of aqueous hydrochloric acid. During this titration, hydrochloric acid is added gradually from a burette until the end-point is reached.

- (i) Write a balanced equation for the reaction of aqueous ammonia with aqueous hydrochloric acid.

.....[1]

- (ii) Suggest a suitable indicator for the titration and indicate the colour change at the end-point.

indicator

colour change[2]

- (iii) 25.0 cm^3 of aqueous ammonia required 22.00 cm^3 of aqueous hydrochloric acid for complete reaction. Calculate the concentration of aqueous ammonia.

[1]

- (iv) During the addition of the first 11.00 cm^3 of hydrochloric acid, the mixture is behaving as a buffer. Explain how the mixture is behaving as a buffer at this stage of the titration.

Include an equation in your answer.

.....

[3]

[Total: 20]

– END OF PAPER –